Customer Churn Prediction Using Tree-Based Models

1. Best Model and Performance

Logistic Regression was the strongest model in the reported test results, with ROC-AUC 0.8448, accuracy 81.17%, precision 68.98%, recall 53.02%, and F1-score 59.96%. It provides the best reported overall discrimination between churn and non-churn customers. However, recall indicates it misses a substantial share of customers who churn, so the operating threshold could be reviewed if identifying more at-risk customers is the business priority.

2. Entropy vs. Gini

Gini and Entropy are two ways a decision tree measures how mixed the classes are when choosing a split. Both produced the same accuracy (71.43%), precision (47.68%), recall (76.87%), and F1-score (58.86%). Their ROC-AUC values were also nearly identical: 0.821443 for Gini and 0.821248 for Entropy. Therefore, the choice made almost no practical difference in these results; Gini had a very small ROC-AUC advantage.

3. Bagging vs. Random Forest and Boosting

Bagging outperformed Random Forest in the reported comparison on accuracy (75.02% vs. 73.51%), precision (52.09% vs. 50.12%), F1-score (61.63% vs. 60.45%), and ROC-AUC (0.8359 vs. 0.8345). Random Forest had slightly higher recall (76.16% vs. 75.44%). The supplied results do not include the final metrics for AdaBoost, Gradient Boosting, and XGBoost, so a reliable conclusion about whether boosting beat or lost to bagging cannot yet be made. The final comparison table should be used to make that conclusion rather than assuming a winner.

4. Top Churn Features and Business Actions

The SHAP summary identified these three most influential features: **tenure (0.1417), TotalCharges (0.0539), and MonthlyCharges (0.0443).** These values represent mean absolute SHAP importance; they show the strength of influence, not whether a feature increases or decreases churn on its own.

Feature	Suggested business action
Tenure	Prioritize early-tenure customers for onboarding calls, service check-ins, and timely support.
TotalCharges	Review customer lifetime and billing patterns together; identify segments with unexpected churn rather than treating high charges as a cause.
MonthlyCharges	Review pricing and plan fit; offer transparent plan comparisons and suitable options to customers whose usage or budget may not match their plan.

Conclusion

Based on the metrics provided, Logistic Regression is the leading model by ROC-AUC. Bagging narrowly exceeds Random Forest on most reported metrics, while Gini and Entropy are effectively tied. Before deployment, compare all tuned boosting models on the same held-out test set and validate that proposed retention actions improve outcomes.